

The Coordination Tax: Light-Speed Delay in a Self-Replicating Probe Swarm Bites Only When Hops Are Long

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Abstract—A swarm of self-replicating probes fills a stellar field star by star, each probe settling a star and launching copies. The standard exploration-timescale model grants every probe perfect, instantaneous knowledge of which stars are already settled, and names finite light-speed as its own future work. We add that light-speed limit: a probe deciding at a star treats a distant star as settled only once the news, travelling at the speed of light, has arrived, so probes race for the same target from stale views and waste trips. Over a 32-seed paired ensemble on identical galaxies, the delay slows the time to fill the field by a median of about 0 percent for powered nearest-neighbour flight, about 30 percent (95 percent bootstrap CI 24 to 38) for nearest-star slingshot flight, and about 51 percent (47 to 54) for maximum-boost slingshot flight; the slingshot penalties are positive in all 32 seeds, hold across coverage fractions from a quarter of the field to all of it, and persist over a four-fold increase in field size. Every case still reaches full coverage, so lag alone leaves no permanent unsettled fraction. The ratio of information delay to travel time across a hop equals the probe speed in units of the speed of light, and it gates whether the delay can bite at all: at the powered cruise speed it is negligible. But because that ratio is the same on every hop, it does not predict the penalty, which instead tracks how non-local each hop is: the two slingshot policies reach comparable speeds, yet the one that reaches past its neighbours to distant stars makes wasted trips about twice as long and pays about seventy percent more. Fast-but-local flight is nearly immune; the cost falls on the long-range hops that made the swarm fast. Under finite light-speed the slingshot speed-up is real, but a meaningful fraction of it is eaten by uncoordinated long-range collisions.

Index Terms—self-replication, interstellar exploration, light-speed delay, multi-agent coordination, swarm dynamics

I. INTRODUCTION

A self-replicating probe does not explore a galaxy alone. It settles a star, builds copies of itself from local material, and sends them on to further stars, so a single launch grows into a front that sweeps outward and fills the reachable field. That a machine can build a copy of itself at all is von Neumann’s result, and a probe of this kind saturating the Galaxy in a few million years is the sharpest form of the Fermi question: if interstellar settlement is possible it should already be complete, so the observed silence is a paradox [1], [2]. How long the filling takes is the exploration-timescale question, and it has a long modelling lineage. Early work framed each interstellar probe as an independent agent acting on priors fixed before launch [3], and treated settlement as a front: a density-dependent diffusion with a travelling-wave

solution [4], or a Monte Carlo migration filling the Galaxy in tens of millions of years [5]. Later simulations made the field concrete, sending probes and subprobes through a defined stellar volume to measure fill times directly [6], [7], and established the seeded Monte Carlo methodology that situates an exploration timescale within a distribution of realizations rather than a single number [8].

The sharpest treatment for self-replicators is the slingshot line. Forgan, Papadogiannakis and Kitching showed that a single probe stealing energy from the motion of the stars it passes, a gravitational slingshot, explores up to two orders of magnitude faster than a probe under its own power [9]. Nicholson and Forgan extended that manoeuvre to self-replicating probes and confirmed the speed-up survives replication, with the surprising result that aiming for the nearest star can beat aiming for the largest boost [10]. The probe speeds that make this regime interesting, from the powered cruise of $3 \times 10^{-5} c$ up toward the 0.1 to 0.2 c of directed-energy or high-velocity concepts [11], [12], are exactly where the next assumption starts to hurt.

That assumption, which Nicholson and Forgan name as a limitation and leave for future work, is that every probe is granted perfect, instantaneous knowledge of which stars are already settled. No probe ever flies to a star another has just claimed, because it always knows. In a real galaxy no such oracle exists: news that a star has been settled can travel no faster than light, so a probe choosing its next target acts on a picture of the field that is, for distant stars, years or centuries out of date. This is the elementary consequence of a finite signal speed, that events have only a partial ordering and a remote event is unknowable until its signal arrives [13].

This paper adds that light-speed limit to the self-replicating slingshot model and measures what it costs. The question is not bookkeeping: it decides whether the celebrated slingshot speed-up survives contact with the finite speed of information, or whether a swarm racing outward on stale maps wastes much of the advantage colliding with itself. The answer, built from a controlled paired-ensemble experiment, is that the tax is real but sharply selective: whether it appears at all is set by the probe’s speed, but how hard it bites is set by how far each hop reaches. Fast-but-local flight escapes almost untouched; fast-and-far flight pays.

II. THE LIGHT-DELAYED BELIEF MODEL

A. The knowledge gate

The change to the model is one gate on knowledge. A settled star is treated as an omnidirectional beacon that begins emitting in the year it is settled. A probe deciding its next target at star f in year Y believes a distant star i is settled only once that beacon's light has had time to arrive:

$$\text{settled}(i) + \frac{\text{dist}(f, i)}{c} \leq Y. \quad (1)$$

Under the perfect-information assumption of the source model, (1) collapses to “settled at any non-negative year,” and the model reproduces our perfect-information baseline bit-for-bit; the light-speed term is what introduces stale views. When two probes both believe an unsettled star is the best target they both fly to it; the loser arrives to find it taken, its trip wasted, and must re-target from the even more stale picture it holds on arrival. This is availability chosen over consistency: rather than wait for a single agreed map that finite light-speed makes impossible, each probe acts now on its local view [13]. Stars carry velocity magnitudes that feed the slingshot boost but are held fixed in position, following the source model, so $\text{dist}(f, i)$ in (1) is static and the beacon geometry does not shift under the probe.

B. The three flight policies

We carry three named policies, following the slingshot studies we extend [9], [10]. *Powered nearest-neighbour* flight moves under the probe's own power at a cruise speed of $3 \times 10^{-5} c$ to the nearest unsettled star. *Nearest-star slingshot* flight takes the same nearest target but gains speed from gravitational slingshots off the stars it passes. *Maximum-boost slingshot* flight instead selects, among the nearest believed-unsettled stars, the target that yields the largest velocity gain, deliberately reaching past near neighbours to distant, fast stars. The three span the axis that matters here: powered flight is slow and local, nearest-star slingshot is fast and local, and maximum-boost slingshot is fast and far. The effective speeds the policies reach in the simulation, which set the lag ratio below and let the reader check the reported penalties, are given with the results in Table II: a powered cruise near 9 km/s against slingshot means near 2,700 km/s (nearest) and 3,900 km/s (maximum-boost).

C. The lag ratio, and what it can and cannot say

How much the gate can matter on a single hop is governed by a dimensionless ratio: the information delay across the hop divided by the travel time across the same hop. For a hop of length d the delay is d/c and the travel time is d/v , so the ratio is

$$\Lambda = \frac{d/c}{d/v} = \frac{v}{c}, \quad (2)$$

the probe's speed in units of c , and it is the *same on every hop*: the length d cancels. A slow probe ($\Lambda \ll 1$) outpaces its own ignorance, so stale views barely arise; at the powered cruise speed $\Lambda \approx 3 \times 10^{-5}$ is negligible. A fast probe carries

a large Λ and routinely decides before the relevant news can reach it. So Λ gates *whether* the delay can bite at all, and it does so correctly: the effect appears only once probes are fast, which is the slingshot regime. But precisely because Λ is hop-length-independent it cannot say *where* the cost lands, and Section III shows the penalty is decided by hop length, not by Λ . We therefore keep Λ for its gating role and demote it as a predictor of the penalty's size.

D. Experimental design and parameters

The field is a box of stars at a uniform density of one star per cubic parsec, the convention of the source model. Each settled star launches two offspring (the replication branching factor); this is what makes the field fill exponentially rather than as a single roving chain, and it sets the swarm density that governs how often two probes contend for one star. Three properties keep the measurement honest. First, knowledge is evaluated only at the decision star at the moment of decision; news a probe's path happens to cross in mid-flight is ignored. This deliberately undercounts what a probe could know, so the measured waste is a conservative upper bound on the true penalty. Second, a probe that loses a race re-targets, but only up to a fixed number of times before it is retired as wasted; this cap bounds pathological bounce chains late in the fill, is bookkeeping rather than physics, and the results are insensitive to it. Third, the whole simulation is a pure, seeded, fixed-step fold: fixing the seed fixes the star field, every arrival, and every target choice, so a run with the light-speed gate and a run without it can be compared on the *identical* galaxy. That paired design is what lets us attribute a difference in fill-time to the delay alone rather than to the luck of the star layout.

The uniform density is easy to over-read, so one clarification. Changing the *value* of the uniform density rescales every inter-star distance by a common factor, which multiplies both the travel time and the light-travel time on every hop equally; Λ and, in the continuum limit, the relative penalty are left unchanged (a pure geometric rescaling of the absolute clock). We confirm this holds in practice: re-running at densities down to the sparser solar-neighbourhood value near 0.14 stars per cubic parsec [14] keeps the median penalty within a few points of its value at unit density (about 27-29% for nearest-star, about 46-52% for maximum-boost), the residual scatter being the fixed timestep, not the density. A *non-uniform*, clumpier distribution is a different matter, and we treat it as a limitation (Section IV-A): by lengthening the long-range hops that carry the penalty it is expected to move the relative penalties, not merely the absolute timescale.

E. Baseline validation

The paired comparison is only as good as its perfect-information baseline, so we check that baseline at two levels. By construction the “instant” regime is the $c \rightarrow \infty$ limit of (1), and it reproduces the default perfect-information fold bit-for-bit. Against Nicholson and Forgan's published model [10] we reproduce their two qualitative findings: gravitational slingshots explore far faster than powered flight, and their

TABLE I

FILL-100% PENALTY FROM LIGHT-SPEED LAG OVER THE 32-SEED PAIRED ENSEMBLE AT 300 STARS: MEDIAN, INTERQUARTILE RANGE, AND A SEEDED PERCENTILE-BOOTSTRAP 95% CONFIDENCE INTERVAL ON THE MEDIAN. THE LAST COLUMN IS A TWO-SIDED SIGN TEST OVER THE PAIRED SEEDS; THE SLINGSHOT PENALTY IS POSITIVE IN ALL 32, WHILE POWERED IS NULL (POSITIVE IN ONLY 6 OF THE 11 SEEDS THAT DIFFER AT ALL).

Flight policy	Median (%)	IQR (%)	95% CI (%)	Sign test
Powered nearest-neighbour	0.0	[0.0, 0.0]	[0.0, 0.0]	1.0
Nearest-star slingshot	30.3	[20.0, 38.5]	[23.5, 37.5]	<10 ⁻⁹
Maximum-boost slingshot	51.4	[46.0, 54.0]	[47.2, 53.7]	<10 ⁻⁹

TABLE II

THE MECHANISM, AS MEDIAN OVER THE 32 LIGHT-SPEED RUNS. EFFECTIVE SPEED IS THE MEAN SPEED AT WHICH PROBES ARE LAUNCHED; $\Lambda_{\text{eff}} = v_{\text{eff}}/c$. HOP LENGTH IS THE MEAN DISTANCE OF A WASTED (LOST-RACE) TRIP AND OF A WINNING (SETTLING) TRIP. THE TWO SLINGSHOT POLICIES DIFFER IN Λ_{eff} BY ABOUT 1.4 $\times$ BUT IN WASTED-HOP LENGTH BY ABOUT 1.85 $\times$, AND THE PENALTY IN TABLE I TRACKS THE LATTER.

Flight policy	v_{eff} (km/s)	Λ_{eff}	Wasted (pc)	Settling (pc)
Powered nearest-neighbour	9	3.0×10^{-5}	1.62	1.62
Nearest-star slingshot	2,734	9.1×10^{-3}	1.44	0.83
Maximum-boost slingshot	3,897	1.3×10^{-2}	2.67	1.44

surprising result that aiming for the nearest star beats aiming for the largest boost on total exploration time. In our fold, on a 400-star field, nearest-star slingshot fills in about 75,000 years against about 305,000 for maximum-boost, while maximum-boost reaches the higher peak speed. We do not reproduce their roughly hundred-fold slingshot speed-up quantitatively: at the fixed timestep the boosted hops are shorter than one step and are quantized, so the measured speed-up is about twenty-fold. Lowering the timestep recovers more of it; the qualitative ordering, on which this paper’s conclusions rest, is timestep-robust.

III. RESULTS: THE COORDINATION TAX

We measured the cost as a paired ensemble: 32 seeded galaxies of 300 stars each, every one filled twice, once with perfect information and once with the light-speed gate of (1), so that the only difference between the two runs of a pair is the delay. The primary quantity is the relative slowdown in the time to settle the entire field, t_{100} , per seed; because t_{100} is a tail statistic dominated by the last few stars, we also report the penalty at earlier coverage fractions and across a range of field sizes. Reporting the distribution across the ensemble rather than a single run follows the seeded Monte Carlo practice of the field [8].

The medians and their spread are given in Table I and shown per seed in Fig. 1. Powered flight pays essentially nothing; nearest-star slingshot loses about a third of its time; maximum-boost slingshot loses about half. The two slingshot penalties are not marginal: each is positive in all 32 seeds, and the bootstrap confidence intervals on the medians (Table I) are well clear of zero and of each other.

The central finding is that the penalty is *not* governed by $\Lambda = v/c$ from (2). Table II makes the case directly. The two slingshot policies reach comparable effective speeds, 2,734 against 3,897 km/s, so their Λ_{eff} differ by only about 1.4 $\times$; yet their penalties differ by about 1.7 $\times$. If Λ set the cost, the faster policy would pay in that same 1.4-to-1 proportion, and the powered policy, though slowest, would still pay something.

Neither holds. What does track the penalty is hop non-locality. A probe flying to the *nearest* star that loses a race has lost almost nothing: the next unsettled star is right beside it, so the wasted trip and the recovery are both short. A probe chasing the *biggest boost* deliberately reaches past its neighbours to a distant star, so when it loses that race both the wasted trip and the re-targeting detour are long. The simulation confirms the mechanism in the hop lengths themselves: the mean wasted trip is 2.67 pc for maximum-boost against 1.44 pc for nearest-star, and its winning trips are nearly three times longer too (Table II). So Λ decides only *whether* lag can bite: powered flight, with $\Lambda \approx 3 \times 10^{-5}$, is immune not because its wasted hops are short (they are not, at 1.62 pc) but because it is far too slow for any race to matter. Given that a policy is fast enough for lag to bite, the *non-locality of its hops* sets how hard. Fig. 4 shows the effect on a single representative galaxy for the nearest-star policy: the light-speed run reaches full settlement at about 110,000 years against 80,000 for the perfect-information run on the same seed, a right-shift of roughly a third visible as a lag in the settlement curve, not a change in its final height.

The penalty is not an artifact of the fragile t_{100} tail. Fig. 2 reports the median penalty at coverage fractions from t_{25} to t_{100} : for nearest-star flight it rises steadily from about 14% at quarter-coverage to about 30% at full coverage, and for maximum-boost it is already about 26% at quarter-coverage and stays near or above 50% thereafter. The shift is present throughout the fill, and grows as the front reaches farther from the origin, exactly as the locality mechanism predicts; it is not a single unlucky last star. Powered flight stays at zero at every fraction.

Nor is it an artifact of the small 300-star box, the concern that matters most for extrapolating toward galactic saturation. Fig. 3 sweeps the system size across a four-fold range: the powered penalty stays near zero, the nearest-star penalty stays near a third, and the maximum-boost penalty holds near or slightly above a half, firming up rather than washing out as the field grows and hops lengthen. The maximum-boost policy scans its nearest neighbours at every launch and so costs $O(N^2)$, which bounds the range we sweep; within that range the penalties are stable, and the trend is toward more penalty at larger scale, not less.

Finally, what does *not* happen: every case still fills the field to full coverage. This is a property of the model, not a result to be discovered. Re-targeting guarantees that a probe which loses a race simply picks another star, and with no loss term the reachable field must eventually close; light-speed lag makes the swarm slower and more wasteful but leaves no permanent unsettled remainder. It is worth stating precisely because it

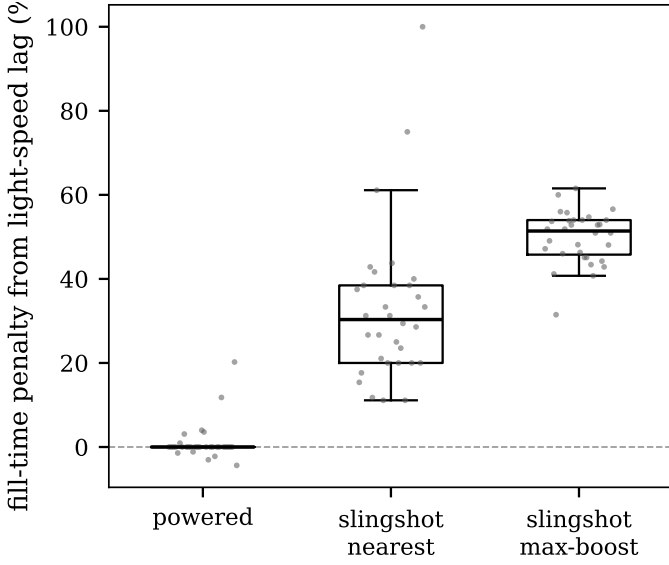


Fig. 1. Distribution of the per-seed fill-100% penalty imposed by light-speed-limited coordination, over the fixed 32-seed ensemble at 300 stars, for the three flight policies (box: interquartile range and median; points: all 32 individual seeds). Powered nearest-neighbour flight pays essentially no penalty, because a lost race is recovered at the star next door, while the long-range maximum-boost policy fills the field materially later, because wasted trips launched from stale views become long detours.

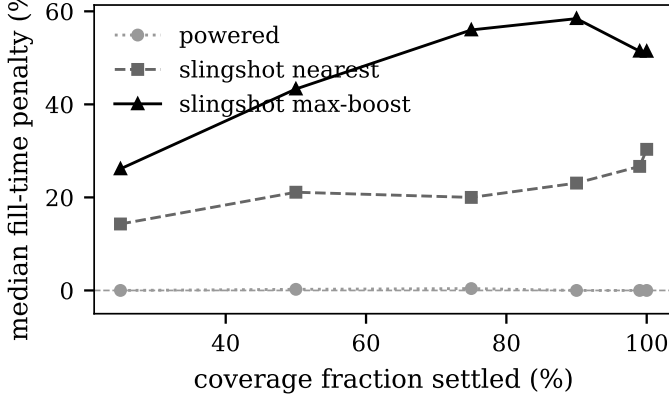


Fig. 2. Median fill-time penalty at coverage fractions from t_{25} to t_{100} , over the 32-seed ensemble, for the three policies. The slowdown is present across the whole fill, not only in the t_{100} tail, and grows as the front reaches farther from the origin; powered flight stays at zero throughout.

marks the boundary of what delay alone can do: a permanent settled fraction below one, an “Aurora” steady state in which some region stays forever empty [12], would require settled sites to *die*, and delay supplies no such term. Lag is a tax on speed, not a barrier to completion.

IV. DISCUSSION

The result refines rather than overturns the slingshot picture. Nicholson and Forgan’s speed-up is real and large [10], but it was computed under perfect coordination, and coordination is not free. Under finite light-speed a meaningful fraction of the advantage, roughly a third for nearest-star slingshots and half for maximum-boost, is spent on trips to stars that turn out to

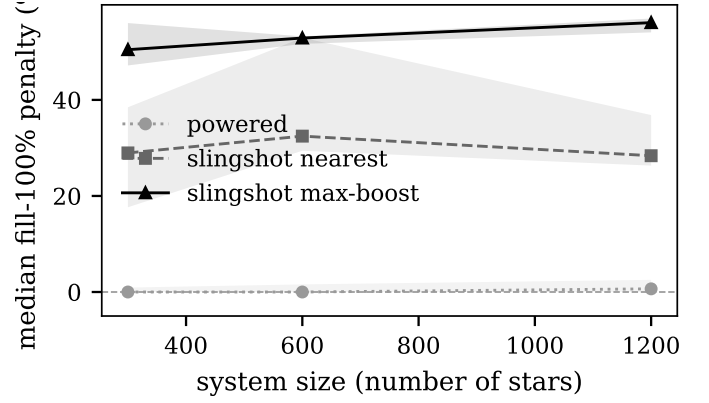


Fig. 3. Median fill-100% penalty versus system size for the three policies (line: median; shaded band: interquartile range over 8 seeds per point). Across a four-fold increase in the number of stars the penalties are stable, holding near 0%, a third, and a half respectively, and firm up rather than wash out. The $O(N^2)$ cost of the maximum-boost scan bounds the range swept.

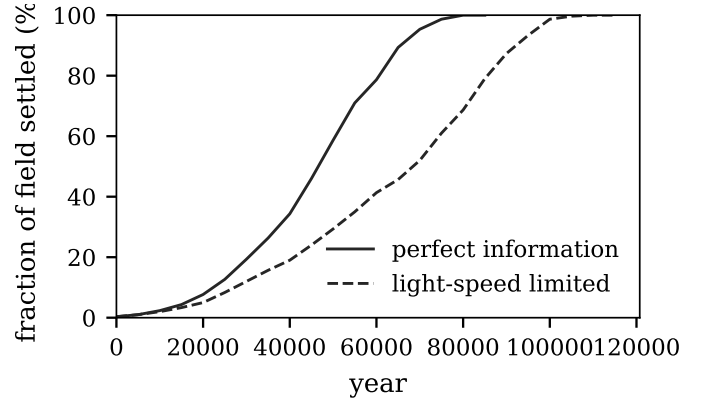


Fig. 4. Fraction of the reachable field settled versus year for the nearest-star slingshot policy on a single representative galaxy, under perfect global information (solid) and under light-speed-limited coordination (dashed). The light-speed curve is shifted later, reaching full settlement at about 110,000 years versus 80,000, a delay attributable to information lag alone; both curves reach the same final height.

be already taken. The more aggressively a policy reaches for distant, high-boost targets, the more of its speed it loses to uncoordinated collisions.

A. Limitations

The measurement is a conservative upper bound in two senses, and both point to future work. Probes here learn only at their decision stars, ignoring news their paths cross in flight, so a two-endpoint knowledge model would recover some of the wasted trips; and probes here share nothing directly, so adding probe-to-probe relay of the settled map would let a swarm partially coordinate rather than merely collide. Both would lower the tax without erasing it, since the fundamental limit, that no probe can know a distant star is taken until its light arrives, is set by physics, not by protocol. Three further simplifications bound the scope. First, the field is uniform. Changing the uniform density only rescales the absolute clock, leaving the relative penalty nearly unchanged (Section II); but

a non-uniform, clumpier distribution is a genuine unknown, and because it would lengthen the long-range hops that carry the penalty we expect it to move the relative penalties upward, making the uniform-field figures conservative for the maximum-boost policy in particular. Second, the maximum-boost scan is $O(N^2)$, which bounded the finite-size sweep at a few thousand stars; the penalties are stable and slightly rising across that range, but the extrapolation to a full galaxy is a trend, not a measurement. Third, the dynamics have no loss term: settled sites never fail, so the field always closes. A settlement-death term of the kind that produces the “Aurora” steady state, in which the settled fraction equilibrates below one [12], is exactly what would be needed to turn lag into a permanent unsettled remainder; delay alone does not.

B. The signature of delayed agreement

The behaviour is, by analogy rather than by derivation, the expected signature of delayed multi-agent agreement. Formal consensus degrades as the ratio of communication delay to decision cadence grows: the standard protocol is stable only while the one-hop delay stays below a bound set by the coupling [15], a result later generalized to switching topologies and link failures [16]. We do not apply that bound quantitatively, because it needs a probe decision cadence the literature does not fix; but the regime is unambiguous. Our probes never reach agreement about the settled map at all: at a mean inter-star spacing of about a parsec the round-trip light-time is several years, far longer than any plausible decision cadence, so they sit deep in the regime where guaranteed agreement is not merely slow but impossible [17], and a system that must nonetheless keep acting is forced to trade a single consistent map for availability [18], which is precisely the choice (1) encodes.

The same progression drove the space teleoperation and networking communities away from real-time control, and it is worth tracing because it fixes where interstellar hops sit. Past about a second of round-trip delay a human operator abandons closed-loop control for move-and-wait [19], and as delay grows the handoff continues through supervisory control to full remote autonomy [20]. Past minutes to hours no continuous end-to-end path exists at all, and communication becomes hop-by-hop store-and-forward, the delay-tolerant regime motivated by the interplanetary case [21] and standardized first as an architecture [22] and now as a deployed protocol [23]. At interstellar hop lengths every one of those faster regimes is left far behind; each probe is effectively an independent colony acting on priors, and the coordination tax is what that independence costs when probes want the same stars.

C. A design lesson

There is a practical corollary. For a real swarm on stale maps, a locally greedy policy is not merely simpler than a boost-maximising one; it is more robust to the delay it cannot avoid, because a lost race costs only the star next door. A designer who cannot buy coordination should prefer locality,

spending the slingshot advantage on speed rather than on reach.

V. CONCLUSION

Adding a finite speed of information to the self-replicating slingshot model turns a convenient assumption into a measurable cost. Over a paired ensemble of identical galaxies, light-speed lag slows the time to fill the field by a median of about 0 percent under powered flight, about 30 percent under nearest-star slingshots, and about 51 percent under maximum-boost slingshots; the slingshot penalties are positive in every one of the 32 seeds, hold across coverage fractions and across a four-fold range of field sizes, and every case still reaches full coverage. The dimensionless ratio $\Lambda = v/c$ gates whether the tax appears at all, and at the powered cruise speed it does not; but because that ratio is the same on every hop it does not predict the penalty’s size. That is decided by how non-local each hop is: fast-but-local flight is nearly immune, while the cost falls on the long-range hops that made the swarm fast in the first place, whose wasted trips we measure to be about twice as long. The slingshot speed-up survives contact with the finite speed of information, but a real swarm pays a coordination tax that a perfect-information model hides, and pays it in proportion to its reach. Two extensions would tighten the estimate without changing its shape: knowledge gained in flight, and probe-to-probe relay of the settled map. Both would lower the tax; neither can remove the physical floor beneath it.

DATA AVAILABILITY

The model is a deterministic, seeded fold; every figure, table, and reported statistic in this paper regenerates from source by running the swarm module’s experiment and figure scripts, and every quantitative claim traces to a cited source in the project’s shared bibliography. The code and its citation metadata are archived with the concept DOI 10.5281/zenodo.21249296.

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